MTRC

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References:

Discrete Mathematical Structure- B. Kolman, R.C. Busby, S.C.Ross & N. Rehman, Pearson

Discrete Mathematical Structure- Lipschutz & Lipson, TMH

Faculty HOD 14/3/2023 DEAN ACADMICS

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

SET THEORY

LECTURE NO:-10

Introduction of Sets

A set is defined as a collection of distinct objects of the same type or class of objects. The purposes of a set are called elements or members of the set. An object can be numbers, alphabets, names, etc.

Examples of sets are:

a. A set of rivers of India.
b. A set of vowels.

We broadly denote a set by the capital letter A, B, C, etc. while the fundamentals of the set by small letter a, b, x, y, etc.

If A is a set, and a is one of the elements of A, then we denote it as a $\in$ A. Here the symbol $\in$ means -"Element of."

Sets Representation:

Sets are represented in two forms:-

a) Roster or tabular form: In this form of representation we list all the elements of the set within braces { } and separate them by commas.

Example: If A= set of all odd numbers less then 10 then in the roster from it can be expressed as A={ 1,3,5,7,9}.

b) Set Builder form: In this form of representation we list the properties fulfilled by all the elements of the set. We note as {x: x satisfies properties P}. and read as 'the set of those entire x such that each x has properties P.'

Example: If B= {2, 4, 8, 16, 32}, then the set builder representation will be: B={x: $x=2^n$, where n $\in$ N and $1 \leq n \geq 5$}

Standard Notations:

x $\in$ A	x belongs to A or x is an element of set A.
x $\notin$ A	x does not belong to set A.

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∅	Empty Set.
U	Universal Set.
N	The set of all natural numbers.
I	The set of all integers.
I_0	The set of all non- zero integers.
I_+	The set of all + ve integers.
C, C_0	The set of all complex, non-zero complex numbers respectively.
Q, Q_0, Q_+	The sets of rational, non- zero rational, +ve rational numbers respectively.
R, R_0, R_+	The set of real, non-zero real, +ve real number respectively.

Cardinality of a Sets:

The total number of unique elements in the set is called the cardinality of the set. The cardinality of the countably infinite set is countably infinite.

Examples:

1. Let P = {k, l, m, n}
The cardinality of the set P is 4.

2. Let A is the set of all non-negative even integers, i.e.
A = {0, 2, 4, 6, 8, 10.....}.

As A is countably infinite set hence the cardinality.

Types of Sets

Sets can be classified into many categories. Some of which are finite, infinite, subset, universal, proper, power, singleton set, etc.

1. Finite Sets: A set is said to be finite if it contains exactly n distinct element where n is a non-negative integer. Here, n is said to be "cardinality of sets." The cardinality of sets is denoted by|A|, # A, card (A) or n (A).

Example:

1. Cardinality of empty set θ is 0 and is denoted by |θ| = 0

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2. Sets of even positive integer is not a finite set.

A set is called a finite set if there is one to one correspondence between the elements in the set and the element in some set n, where n is a natural number and n is the cardinality of the set. Finite Sets are also called numerable sets. n is termed as the cardinality of sets or a cardinal number of sets.

2. Infinite Sets: A set which is not finite is called as Infinite Sets.

Countable Infinite: If there is one to one correspondence between the elements in set and element in N. A countably infinite set is also known as Denumerable. A set that is either finite or denumerable is known as countable. A set which is not countable is known as Uncountable. The set of a non-negative even integer is countable Infinite.

Uncountable Infinite: A set which is not countable is called Uncountable Infinite Set or non-denumerable set or simply Uncountable.

Example: Set R of all +ve real numbers less than 1 that can be represented by the decimal form 0. a_1,a_2,a_3..... Where a_1 is an integer such that $0 \leq a_i \leq 9$.

3. Subsets: If every element in a set A is also an element of a set B, then A is called a subset of B. It can be denoted as $A \subseteq B$. Here B is called Superset of A.

Example: If A= {1, 2} and B= {4, 2, 1} the A is the subset of B or $A \subseteq B$.

Properties of Subsets:

1. Every set is a subset of itself.
2. The Null Set i.e.$\varnothing$ is a subset of every set.
3. If A is a subset of B and B is a subset of C, then A will be the subset of C. If $A \subset B$ and $B \subset C \Longrightarrow A \subset C$
4. A finite set having n elements has 2^n subsets.

4. Proper Subset: If A is a subset of B and $A \neq B$ then A is said to be a proper subset of B. If A is a proper subset of B then B is not a subset of A, i.e., there is at least one element in B which is not in A.

Example:

(i) Let A = {2, 3, 4}
B = {2, 3, 4, 5}

A is a proper subset of B.

(ii) The null $\varnothing$ is a proper subset of every set.

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5. **Improper Subset:** If A is a subset of B and A = B, then A is said to be an improper subset of B.

Example

(i) A = {2, 3, 4}, B = {2, 3, 4}

A is an improper subset of B.

(ii) Every set is an improper subset of itself.

6. **Universal Set:** If all the sets under investigations are subsets of a fixed set U, then the set U is called Universal Set.

Example: In the human population studies the universal set consists of all the people in the world.

7. **Null Set or Empty Set:** A set having no elements is called a Null set or void set. It is denoted by $\varnothing$.

8. Singleton Set: It contains only one element. It is denoted by {s}.

Example: $S = \{x | x \in N, 7<x<9\} = \{8\}$

9. **Equal Sets:** Two sets A and B are said to be equal and written as A = B if both have the same elements. Therefore, every element which belongs to A is also an element of the set B and every element which belongs to the set B is also an element of the set A.

1. $A = B \iff \{x \in A \iff x \in B\}$.

If there is some element in set A that does not belong to set B or vice versa then $A \neq B$, i.e., A is not equal to B.

10. **Equivalent Sets:** If the cardinalities of two sets are equal, they are called equivalent sets.

Example: If A= {1, 2, 6} and B= {16, 17, 22}, they are equivalent as cardinality of A is equal to the cardinality of B. i.e. $|A|=|B|=3$

11. **Disjoint Sets:** Two sets A and B are said to be disjoint if no element of A is in B and no element of B is in A.

Example:

R = {a, b, c}
S = {k, p, m}

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R and S are disjoint sets.

12. **Power Sets:** The power of any given set A is the set of all subsets of A and is denoted by **P (A)**. If A has n elements, then **P (A)** has 2^n elements.

Example: A = {1, 2, 3}
P (A) = {∅ , {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}}.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

SET OPERATIONS

LECTURE NO:-11

Operations on Sets

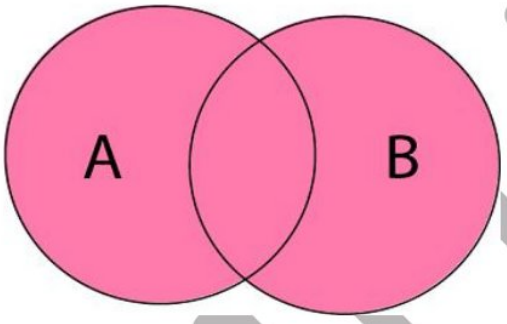
The basic set operations are:

1. **Union of Sets:** Union of Sets A and B is defined to be the set of all those elements which belong to A or B or both and is denoted by A ∪ B.

1. $A \cup B = \{x: x \in A \text{ or } x \in B\}$

Example: Let A = {1, 2, 3}, B= {3, 4, 5, 6}
A ∪ B = {1, 2, 3, 4, 5, 6}.

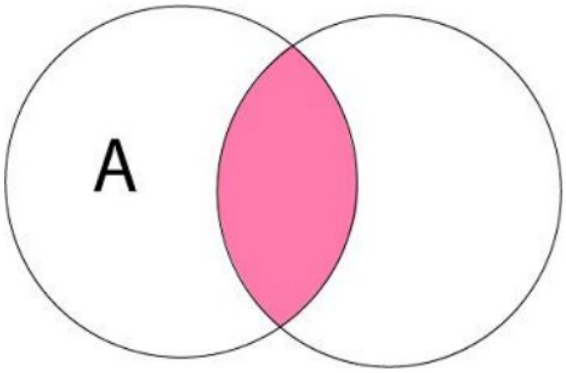
A ∪ B = {1, 2, 3, 4, 5, 6}.



2. **Intersection of Sets:** Intersection of two sets A and B is the set of all those elements which belong to both A and B and is denoted by A ∩ B.

1. $A \cap B = \{x: x \in A \text{ and } x \in B\}$

Example: Let A = {11, 12, 13}, B = {13, 14, 15}
A ∩ B = {13}.



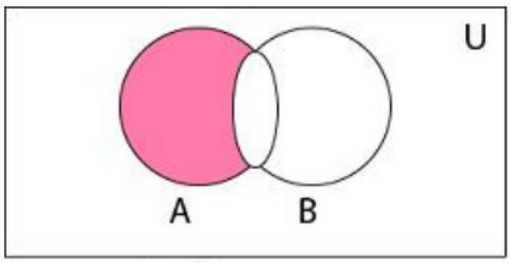
3. **Difference of Sets:** The difference of two sets A and B is a set of all those elements which belongs to A but do not belong to B and is denoted by A - B.

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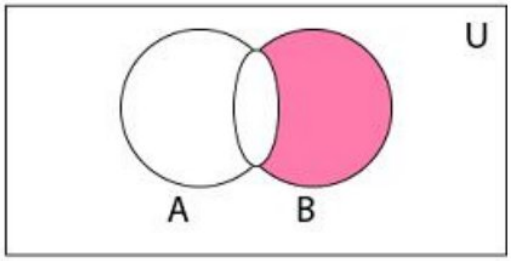
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1. $A - B = \{x: x \in A \text{ and } x \notin B\}$

Example: Let A = {1, 2, 3, 4} and B = {3, 4, 5, 6} then A - B = {3, 4} and B - A = {5, 6}



A - B

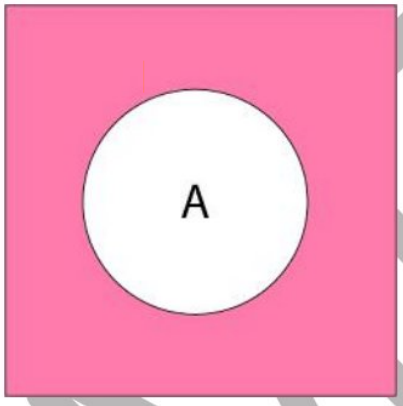


B - A

4. **Complement of a Set:** The Complement of a Set A is a set of all those elements of the universal set which do not belong to A and is denoted by A^c.

$A^c = U - A = \{x: x \in U \text{ and } x \notin A\} = \{x: x \notin A\}$

Example: Let U is the set of all natural numbers.
A = {1, 2, 3}
A^c = {all natural numbers except 1, 2, and 3}.

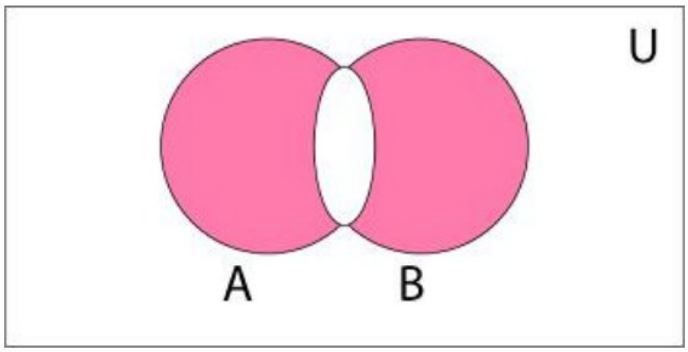


5. **Symmetric Difference of Sets:** The symmetric difference of two sets A and B is the set containing all the elements that are in A or B but not in both and is denoted by $A \oplus B$ i.e.

1. $A \oplus B = (A \cup B) - (A \cap B)$

Example: Let A = {a, b, c, d}
B = {a, b, l, m}
$A \oplus B$ = {c, d, l, m}

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Algebra of Sets

Sets under the operations of union, intersection, and complement satisfy various laws (identities) which are listed in Table 1.

Table: Law of Algebra of Sets

Idempotent Laws	(a) $A \cup A = A$	(b) $A \cap A = A$
Associative Laws	(a) $(A \cup B) \cup C = A \cup (B \cup C)$	(b) $(A \cap B) \cap C = A \cap (B \cap C)$
Commutative Laws	(a) $A \cup B = B \cup A$	(b) $A \cap B = B \cap A$
Distributive Laws	(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
De Morgan's Laws	(a) $(A \cup B)^c = A^c \cap B^c$	(b) $(A \cap B)^c = A^c \cup B^c$
Identity Laws	(a) $A \cup \varnothing = A$ (b) $A \cup U = U$	(c) $A \cap U = A$ (d) $A \cap \varnothing = \varnothing$
Complement Laws	(a) $A \cup A^c = U$ (b) $A \cap A^c = \varnothing$	(c) $U^c = \varnothing$ (d) $\varnothing^c = U$

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Involution Law	(a) $(A^c)^c = A$	

Partitioning of a Set

Partition of a set, say S, is a collection of n disjoint subsets, say P1, P1, ... Pn that satisfies the following three conditions −

- Pi does not contain the empty set.
  [Pi ≠ { ∅ } for all 0 < i ≤ n]
- The union of the subsets must equal the entire original set.
  [P1 ∪ P2 ∪ ... ∪ Pn = S]
- The intersection of any two distinct sets is empty.
  [Pa ∩ Pb = { ∅ }, for a ≠ b where n ≥ a, b ≥ 0]

Example

Let S = { a, b, c, d, e, f, g, h }

One probable partitioning is { a }, { b, c, d }, { e, f, g, h }

Another probable partitioning is { a, b }, { c, d }, { e, f, g, h }

Bell Numbers

Bell numbers give the count of the number of ways to partition a set. They are denoted by Bn where n is the cardinality of the set.

Example −

Let S = { 1, 2, 3}, n = |S| = 3

The alternate partitions are −

1. ∅ , { 1, 2, 3 }
2. { 1 } , { 2, 3 }
3. { 1, 2 } , { 3 }
4. { 1, 3 } , { 2 }
5. { 1 } , { 2 } , { 3 }

Hence B3 = 5

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

Problems on Sets

LECTURE NO:-12

1. Let A and B be two finite sets such that n(A) = 20, n(B) = 28 and n(A ∪ B) = 36, find n(A ∩ B).

Solution:

Using the formula n(A ∪ B) = n(A) + n(B) - n(A ∩ B).

then n(A ∩ B) = n(A) + n(B) - n(A ∪ B)

= 20 + 28 - 36

= 48 - 36

= 12

2. If n(A - B) = 18, n(A ∪ B) = 70 and n(A ∩ B) = 25, then find n(B).

Solution:

Using the formula n(A ∪ B) = n(A - B) + n(A ∩ B) + n(B - A)

70 = 18 + 25 + n(B - A)

70 = 43 + n(B - A)

n(B - A) = 70 - 43

n(B - A) = 27

Now n(B) = n(A ∩ B) + n(B - A)

= 25 + 27

= 52

Different types on word problems on sets:

3. In a group of 60 people, 27 like cold drinks and 42 like hot drinks and each person likes at least one of the two drinks. How many like both coffee and tea?

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Solution:

Let A = Set of people who like cold drinks.

B = Set of people who like hot drinks.

Given

$(A \cup B) = 60$ $n(A) = 27$ $n(B) = 42$ then;

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 27 + 42 - 60$

$= 69 - 60 = 9$

$= 9$

Therefore, 9 people like both tea and coffee.

4. There are 35 students in art class and 57 students in dance class. Find the number of students who are either in art class or in dance class.

- When two classes meet at different hours and 12 students are enrolled in both activities.
- When two classes meet at the same hour.

Solution:

$n(A) = 35$, $n(B) = 57$, $n(A \cap B) = 12$

(Let A be the set of students in art class.
B be the set of students in dance class.)

(i) When 2 classes meet at different hours $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 35 + 57 - 12$

$= 92 - 12$

$= 80$

(ii) When two classes meet at the same hour, $A \cap B = \varnothing$ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= n(A) + n(B)$

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= 35 + 57

= 92

5. In a group of 100 persons, 72 people can speak English and 43 can speak French. How many can speak English only? How many can speak French only and how many can speak both English and French?

Solution:

Let A be the set of people who speak English.

B be the set of people who speak French.

A - B be the set of people who speak English and not French.

B - A be the set of people who speak French and not English.

A ∩ B be the set of people who speak both French and English.

Given,

n(A) = 72 n(B) = 43 n(A ∪ B) = 100

Now, n(A ∩ B) = n(A) + n(B) - n(A ∪ B)

= 72 + 43 - 100

= 115 - 100

= 15

Therefore, Number of persons who speak both French and English = 15

n(A) = n(A - B) + n(A ∩ B)

⇒ n(A - B) = n(A) - n(A ∩ B)

= 72 - 15

= 57

and n(B - A) = n(B) - n(A ∩ B)

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= 43 - 15 = 28

Therefore, Number of people speaking English only = 57

Number of people speaking French only = 28

Word problems on sets using the different properties (Union & Intersection):

6. In a competition, a school awarded medals in different categories. 36 medals in dance, 12 medals in dramatics and 18 medals in music. If these medals went to a total of 45 persons and only 4 persons got medals in all the three categories, how many received medals in exactly two of these categories?

Solution:

Let A = set of persons who got medals in dance.

B = set of persons who got medals in dramatics.

C = set of persons who got medals in music.

Given,

$n(A) = 36$ $n(B) = 12$ $n(C) = 18$

$n(A \cup B \cup C) = 45$ $n(A \cap B \cap C) = 4$

We know that number of elements belonging to exactly two of the three sets A, B, C

$= n(A \cap B) + n(B \cap C) + n(A \cap C) - 3n(A \cap B \cap C)$

$= n(A \cap B) + n(B \cap C) + n(A \cap C) - 3 \times 4$ .....(i)

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

Therefore, $n(A \cap B) + n(B \cap C) + n(A \cap C) = n(A) + n(B) + n(C) + n(A \cap B \cap C) - n(A \cup B \cup C)$

From (i) required number

$= n(A) + n(B) + n(C) + n(A \cap B \cap C) - n(A \cup B \cup C) - 12$

$= 36 + 12 + 18 + 4 - 45 - 12$

$= 70 - 57$

$= 13$

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7. Each student in a class of 40 plays at least one indoor game chess, carrom and scrabble. 18 play chess, 20 play scrabble and 27 play carrom. 7 play chess and scrabble, 12 play scrabble and carrom and 4 play chess, carrom and scrabble. Find the number of students who play (i) chess and carrom. (ii) chess, carrom but not scrabble.

Solution:

Let A be the set of students who play chess

B be the set of students who play scrabble

C be the set of students who play carrom

Therefore, We are given $n(A \cup B \cup C) = 40$,

$n(A) = 18$, $n(B) = 20$ $n(C) = 27$,

$n(A \cap B) = 7$, $n(C \cap B) = 12$ $n(A \cap B \cap C) = 4$

We have

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

Therefore, $40 = 18 + 20 + 27 - 7 - 12 - n(C \cap A) + 4$

$40 = 69 – 19 - n(C \cap A)$

$40 = 50 - n(C \cap A)$ $n(C \cap A) = 50 - 40$

$n(C \cap A) = 10$

Therefore, Number of students who play chess and carrom are 10.

Also, number of students who play chess, carrom and not scrabble.

$= n(C \cap A) - n(A \cap B \cap C)$

$= 10 – 4$

$= 6$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

INTRODUCTION AND TYPES OF RELATIONS

LECTURE NO:-13

Introduction and types of Relations

Relation or Binary relation R from set A to B is a subset of AxB which can be defined as aRb ↔ (a,b) € R ↔ R(a,b).
A Binary relation R on a single set A is defined as a subset of AxA. For two distinct set, A and B with cardinalities m and n, the maximum cardinality of the relation R from A to B is mn.

Domain and Range:
if there are two sets A and B and Relation from A to B is R(a,b), then domain is defined as the set { a | (a,b) € R for some b in B} and Range is defined as the set {b | (a,b) € R for some a in A}.

Types of Relation:

1. ordered pairs (a,b). For Irreflexive relation, no (a,a) holds for every element a in R. It**Empty Relation:** A relation R on a set A is called Empty if the set A is empty set.
2. **Full Relation:** A binary relation R on a set A and B is called full if AXB.
3. **Reflexive Relation:** A relation R on a set A is called reflexive if (a,a) € R holds for every element a € A .i.e. if set A = {a,b} then R = {(a,a), (b,b)} is reflexive relation.
4. **Irreflexive relation :** A relation R on a set A is called reflexive if no (a,a) € R holds for every element a € A.i.e. if set A = {a,b} then R = {(a,b), (b,a)} is irreflexive relation.
5. **Symmetric Relation:** A relation R on a set A is called symmetric if (b,a) € R holds when (a,b) € R.i.e. The relation R={(4,5),(5,4),(6,5),(5,6)} on set A={4,5,6} is symmetric.
6. **AntiSymmetric Relation:** A relation R on a set A is called antisymmetric if (a,b)€ R and (b,a) € R then a = b is called antisymmetric.i.e. The relation R = {(a,b)→ R|a ≤ b} is anti-symmetric since a ≤ b and b ≤ a implies a = b.
7. **Transitive Relation:** A relation R on a set A is called transitive if (a,b) € R and (b,c) € R then (a,c) € R for all a,b,c € A.i.e. Relation R={(1,2),(2,3),(1,3)} on set A={1,2,3} is transitive.
8. **Equivalence Relation:** A relation is an Equivalence Relation if it is reflexive, symmetric, and transitive. i.e. relation R={(1,1),(2,2),(3,3),(1,2),(2,1),(2,3),(3,2),(1,3),(3,1)} on set A={1,2,3} is equivalence relation as it is reflexive, symmetric, and transitive.
9. **Asymmetric relation:** Asymmetric relation is opposite of symmetric relation. A relation R on a set A is called asymmetric if no (b,a) € R when (a,b) € R.

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Important Points:

1. Symmetric and anti-symmetric relations are not opposite because a relation R can contain both the properties or may not.

2. A relation is asymmetric if and only if it is both anti-symmetric and irreflexive.

3. Number of different relation from a set with n elements to a set with m elements is 2^{mn}

Ex:

if R ={r_1, r_2, r_3.....r_n} and S ={s_1, s_2, s_3.....s_m}
then Cartesian product of R and S is:
R X S = {(r_1, s_1), (r_1, s_2), (r_1, s_3)....., (r_1, s_n),
(r_2, s_1), (r_2, s_2), (r_2, s_3)....., (r_2, s_n),
.....
(r_n, s_1),(r_n, s_2), (r_n, s_3),....., (r_n, s_n)}

This set of ordered pairs contains mn pairs.
Now these pairs can be present in R X S or can be absent.
So total number of possible relation = 2^{mn}

4. Number of Reflexive Relations on a set with n elements : $2^{n(n-1)}$.

A relation has ordered pairs (a,b). Now a can be chosen in n ways and same for b. So set of ordered pairs contains n^2 pairs. Now for a reflexive relation, (a,a) must be present in these ordered pairs. And there will be total n pairs of (a,a), so number of ordered pairs will be n^2-n pairs. So total number of reflexive relations is equal to $2^{n(n-1)}$.

5. Number of Symmetric Relations on a set with n elements : $2^{n(n+1)/2}$.

A relation has ordered pairs (a,b). Now for a symmetric relation, if (a,b) is present in R, then (b,a) must be present in R.
In Matrix form, if a_{12} is present in relation, then a_{21} is also present in relation and As we know reflexive relation is part of symmetric relation.
So from total n^2 pairs, only n(n+1)/2 pairs will be chosen for symmetric relation. So total number of symmetric relation will be $2^{n(n+1)/2}$.

6. Number of Anti-Symmetric Relations on a set with n elements: 2^n $3^{n(n-1)/2}$.

A relation has ordered pairs (a,b). For anti-symmetric relation, if (a,b) and (b,a) is present in relation R, then a = b.(That means a is in relation with itself for any a).
So for (a,a), total number of ordered pairs = n and total number of relation = 2^n.
if (a,b) and (b,a) both are not present in relation or Either (a,b) or (b,a) is not present in relation. So there are three possibilities and total number of ordered pairs for this condition is n(n-1)/2. (selecting a pair is same as selecting the two numbers from n without repetition) As we have to find number of ordered pairs where a ≠ b. it is like opposite of symmetric relation means total number of ordered pairs = (n^2) – symmetric ordered pairs(n(n+1)/2) = n(n-1)/2. So, total number of relation is $3^{n(n-1)/2}$.
So total number of anti-symmetric relation is $2^n.3^{n(n-1)/2}$.

7. Number of Asymmetric Relations on a set with n elements : $3^{n(n-1)/2}$.

In Asymmetric Relations, element a can not be in relation with itself. (i.e. there is no aRa ∀ a∈A relation.) And Then it is same as Anti-Symmetric Relations.(i.e. you have

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three choice for pairs (a,b) (b,a)). Therefore there are $3^{n(n-1)/2}$ Asymmetric Relations possible.

8. Irreflexive Relations on a set with n elements : $2^{n(n-1)}$.

A relation has or is also opposite of reflexive relation.

Now for a Irreflexive relation, (a,a) must not be present in these ordered pairs means total n pairs of (a,a) is not present in R, So number of ordered pairs will be n^2-n pairs. So total number of reflexive relations is equal to $2^{n(n-1)}$.

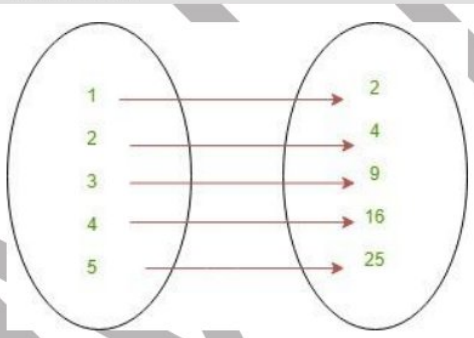
9. Reflexive and symmetric Relations on a set with n elements : $2^{n(n-1)/2}$.

A relation has ordered pairs (a,b). Reflexive and symmetric Relations means (a,a) is included in R and (a,b)(b,a) pairs can be included or not. (In Symmetric relation for pair (a,b)(b,a) (considered as a pair). whether it is included in relation or not) So total number of Reflexive and symmetric Relations is $2^{n(n-1)/2}$.

Representing Relations

Relations are represented using ordered pairs, matrix and digraphs:

1. **Ordered Pairs –**
   In this set of ordered pairs of x and y are used to represent relation. In this corresponding values of x and y are represented using parenthesis.
2. Example: {(1, 1), (2, 4), (3, 9), (4, 16), (5, 25)}
3. This represent square of a number which means if x=1 then y = x*x = 1 and so on.



4. **Representing using Matrix –**
   In this zero-one is used to represent the relationship that exists between two sets. In this if a element is present then it is represented by 1 else it is represented by 0. In this method it is easy to judge if a relation is reflexive, symmetric or transitive just by looking at the matrix.
5. Suppose R is a relation from X={x1, x2, ...xn} to Y={y1, y2....yn}
6. It is represented by :-
7. M[i, j]={1, if (Xi, Yj) belongs to R
8. 0, if (Xi, Yj) does not belong to R}

If A={1, 2, 3} and B={1, 2} and Relation R is
R = {(2, 1), (3, 1), (3, 2)}
then all corresponding value of Relation will be represented by “1” else “0”.

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1. It is represented as:

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

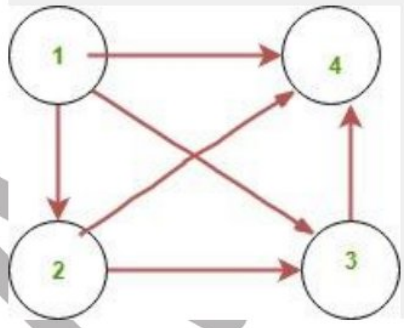
It's corresponding possible relations are:

$$\begin{bmatrix} (1,1) & (1,2) \\ (2,1) & (2,2) \\ (3,1) & (3,2) \end{bmatrix}$$

2. **Digraph –**
   A digraph is known was directed graph. It consists of set 'V' of vertices and with the edges 'E'. Here E is represented by ordered pair of Vertices.
   In the edge (a, b), a is the initial vertex and b is the final vertex.
   If edge is (a, a) then this is regarded as loop.
   Example: Suppose we have relation forming

   R = {(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}

   This relation is represented using digraph as:



Representations of Matrices and Graphs in Relations

Combining Relation:

Suppose R is a relation from set A to B and S is a relation from set B to C, the combination of both the relations is the relation which consists of ordered pairs (a,c) where a Є A and c Є C and there exist an element b Є B for which (a,b) Є R and (b,c) Є S. This is represented as RoS.

Inverse Relation:

A relation R is defined as (a,b) Є R from set A to set B, then the inverse relation is defined as (b,a) Є R from set B to set A. Inverse Relation is represented as R^{-1}

$R^{-1} = \{(b,a) \mid (a,b) \in R\}$.

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Complementary Relation:

Let R be a relation from set A to B, then the complementary Relation is defined as- {(a,b) } where (a,b) is not Є R.

Represenation of Relations:

Relations can be represented as- Matrices and Directed graphs.

Relation as Matrices:

A relation R is defined as from set A to set B,then the matrix representation of relation is M_R= $[m_{ij}]$ where

m_{ij} = { 1, if (a,b) Є R

0, if (a,b) Є R }

Properties:

1. A relation R is reflexive if the matrix diagonal elements are 1.
2. A relation R is irreflexive if the matrix diagonal elements are 0.
3. A relation R is symmetric if the transpose of relation matrix is equal to its original relation matrix. i.e. $M_R = (M_R)^T$.
   A relation R is antisymmetric if either $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.
4. A relation follows join property i.e. the join of matrix M1 and M2 is M1 V M2 which is represented as R1 U R2 in terms of relation.
5. A relation follows meet property i.r. the meet of matrix M1 and M2 is M1 ^ M2 which is represented as R1 Λ R2 in terms of relation.

Relations as Directed graphs:

A directed graph consists of nodes or vertices connected by directed edges or arcs. Let R is relation from set A to set B defined as (a,b) Є R, then in directed graph-it is represented as edge(an arrow from a to b) between (a,b).

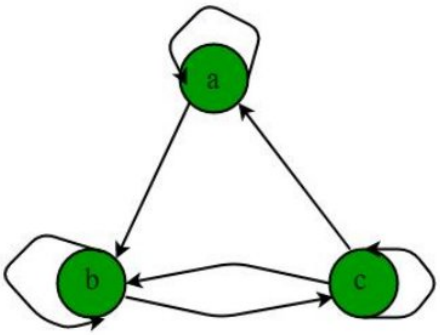
Properties:

1. A relation R is reflexive if there is loop at every node of directed graph.
2. A relation R is irreflexive if there is no loop at any node of directed graphs.
3. A relation R is symmetric if for every edge between distinct nodes, an edge is always present in opposite direction.
4. A relation R is asymmetric if there are never two edges in opposite direction between distinct nodes.
5. A relation R is transitive if there is an edge from a to b and b to c, then there is always an edge from a to c.

Example:

The directed graph of relation R = {(a,a),(a,b),(b,b),(b,c),(c,c),(c,b),(c,a)} is represented as :

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Since, there is loop at every node,it is reflexive but it is neither symmetric nor antisymmetric as there is an edge from a to b but no opposite edge from b to a and also directed edge from b to c in both directions. R is not transitive as there is an edge from a to b and b to c but no edge from a to c.

Example: Let A = {1, 2, 3, 4} and R = {(1, 1), (1, 3), (2, 2), (2, 4), (3, 1), (3, 3), (4, 2), (4, 4)}.

Show that R is an Equivalence Relation.

Solution:

Reflexive: Relation R is reflexive as (1, 1), (2, 2), (3, 3) and (4, 4) $\in$ R.

Symmetric: Relation R is symmetric because whenever (a, b) $\in$ R, (b, a) also belongs to R.

Example: (2, 4) $\in$ R $\Longrightarrow$(4, 2) $\in$ R.

Transitive: Relation R is transitive because whenever (a, b) and (b, c) belongs to R, (a, c) also belongs to R.

Example: (3, 1) $\in$ R and (1, 3) $\in$ R $\Longrightarrow$(3, 3) $\in$ R.

So, as R is reflexive, symmetric and transitive, hence, R is an Equivalence Relation.

Example: A = {1, 2, 3}

R_1 = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)}

R_2 = {(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)}

$R_1 \cup R_2$= {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)}

Hence, Reflexive or Symmetric are Equivalence Relation but transitive may or may not be an equivalence relation.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

FUNCTION & TYPES OF FUNCTIONS

LECTURE NO:-14

Function & Types of Functions

A ***function*** is a rule that assigns each input exactly one output. We call the output the ***image*** of the input. The set of all inputs for a function is called the ***domain***. The set of all allowable outputs is called the ***codomain***. We would write f:X→Y to describe a function with name f, domain X and codomain Y. This does not tell us *which* function ff is though. To define the function, we must describe the rule. This is often done by giving a formula to compute the output for any input (although this is certainly not the only way to describe the rule).

For example, consider the function f:N→N defined by $f(x)=x^2+3$. Here the domain and codomain are the same set (the natural numbers). The rule is: take your input, multiply it by itself and add 3. This works because we can apply this rule to every natural number (every element of the domain) and the result is always a natural number (an element of the codomain). Notice though that not every natural number is actually an output (there is no way to get 0, 1, 2, 5, etc.). The set of natural numbers that *are* outputs is called the ***range*** of the function (in this case, the range is {3,4,7,12,19,28,…}, all the natural numbers that are 3 more than a perfect square).

The key thing that makes a rule a *function* is that there is *exactly one* output for each input. That is, it is important that the rule be a good rule. What output do we assign to the input 7? There can only be one answer for any particular function.

The following are all examples of functions:

1. f:Z→Z defined by f(n)=3n. The domain and codomain are both the set of integers. However, the range is only the set of integer multiples of 3.
2. g:{1,2,3}→{a,b,c} defined by g(1)=c,, g(2)=a and g(3)=a. The domain is the set {1,2,3}, the codomain is the set {a,b,c} and the range is the set {a,c}. Note that g(2) and g(3) are the same element of the codomain. This is okay since each element in the domain still has only one output.
3. h:{1,2,3,4}→N defined by the table:

x	1	2	3	4
h(x)	3	6	9	12

4. Here the domain is the finite set {1,2,3,4} and to codomain is the set of natural numbers, N.N. At first you might think this function is the same as ff defined above. It is absolutely not. Even though the rule is the same,

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the domain and codomain are different, so these are two different functions.

Example

Just because you can describe a rule in the same way you would write a function, does not mean that the rule is a function. The following are NOT functions.

1. f:N→N defined by f(n)=n2. The reason this is not a function is because not every input has an output. Where does ff send 3? The rule says that f(3)=32, but 32 is not an element of the codomain.
2. Consider the rule that matches each person to their phone number. If you think of the set of people as the domain and the set of phone numbers as the codomain, then this is not a function, since some people have two phone numbers. Switching the domain and codomain sets doesn't help either, since some phone numbers belong to multiple people (assuming some households still have landlines when you are reading this).

Injective / One-to-one function

A function f:A→B is injective or one-to-one function if for every b∈B, there exists at most one a∈A such that f(s)=t.
This means a function **f** is injective if a1≠a2 implies f(a1)≠f(a2).

for x,y belongs to A
if x!=y then
f(x)!=f(y)
or if f(x)=f(y)
then x=y

Example

- f:N→N,f(x)=5x is injective.
- f:N→N,f(x)=x^2 is injective.
- f:R→R,f(x)=x^2 is not injective as $(-x)^2=x^2$

Surjective / Onto function

A function f:A→B is surjective (onto) if the image of f equals its range. Equivalently, for every b∈B, there exists some a∈A such that f(a)=b. This means that for any y in B, there exists some x in A such that y=f(x).

Example

- f:N→N,f(x)=x+2 is surjective.
- f:R→R,f(x)=x^2 is not surjective since we cannot find a real number whose square is negative.

Bijective / One-to-one Correspondent

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A function f:A→B is bijective or one-to-one correspondent if and only if **f** is both injective and surjective.

Problem

Prove that a function f:R→R defined by f(x)=2x–3f is a bijective function.

Explanation – We have to prove this function is both injective and surjective.

If f(x1)=f(x2), then 2x1–3=2x2–3and it implies that x1=x2.

Hence, f is **injective**.

Here, 2x–3=y
So, x=(y+5)/3 which belongs to R and f(x)=y

Hence, f is **surjective**.

Since **f** is both **surjective** and **injective**, we can say **f** is **bijective**.

Inverse of a Function

The **inverse** of a one-to-one corresponding function f:A→B, is the function g:B→A, holding the following property –
f(x)=y⇔g(y)=x

The function f is called **invertible**, if its inverse function g exists.

Example

- A Function f:Z→Z,f(x)=x+5, is invertible since it has the inverse function g:Z→Z,g(x)=x−5.
- A Function f:Z→Z,f(x)=x^2 is not invertiable since this is not one-to-one as (−x)2=x2(−x)2=x2.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

COMPOSITION OF FUNCTION & RECURSIVELY DEFINED FUNCTIONS

LECTURE NO:-15

Composition of Function & Recursively Defined Functions

Two functions f:A→B and g:B→C can be composed to give a composition gof. This is a function from A to C defined by (gof)(x)=g(f(x))

Example

Let f(x)=x+2 and g(x)=2x+1, find (fog)(x) and (gof)(x).

Solution

(fog)(x)=f(g(x))=f(2x+1)=2x+1+2=2x+3
(gof)(x)=g(f(x))=g(x+2)=2(x+2)+1=2x+5
Hence, (fog)(x)≠(gof)(x)

Some Facts about Composition

- If f and g are one-to-one then the function (gof) is also one-to-one.
- If f and g are onto then the function (gof) is also onto.
- Composition always holds associative property but does not hold commutative property.

Compositions of Functions

Consider functions, f: A → B and g: B → C. The composition of f with g is a function from A into C defined by (gof) (x) = g [f(x)] and is defined by gof.

To find the composition of f and g, first find the image of x under f and then find the image of f (x) under g.

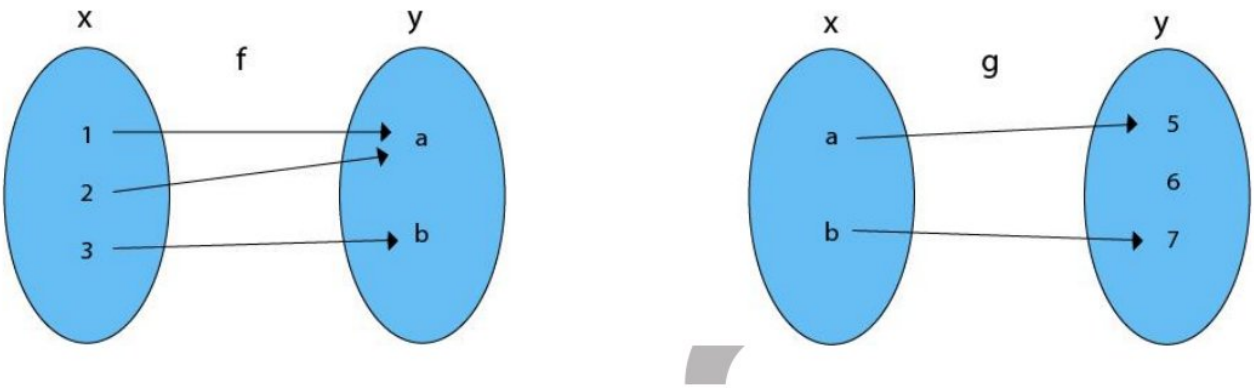
Example1

1. Let X = {1, 2, 3}
2. Y = {a, b}
3. Z = {5, 6, 7}.

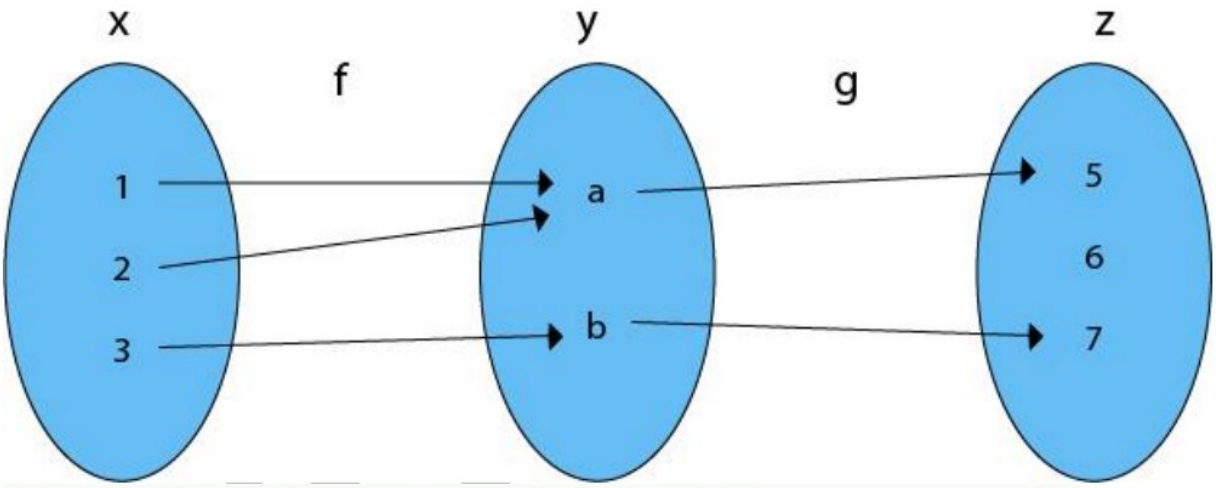
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Consider the function f = {(1, a), (2, a), (3, b)} and g = {(a, 5), (b, 7)} as in figure. Find the composition of gof.



Solution: The composition function gof is shown in fig:



(gof) (1) = g [f (1)] = g (a) = 5, (gof) (2) = g [f (2)] = g (a) = 5
(gof) (3) = g [f (3)] = g (b) = 7.

Example2: Consider f, g and h, all functions on the integers, by f (n) =n^2, g (n) = n + 1 and h (n) = n - 1.

Determine **(i)** hofog **(ii)** gofoh **(iii)** fogoh.

Solution:

(i) hofog (n) = n + 1,
hofog (n + 1) = $(n+1)^2$
h [$(n+1)^2$] = $(n+1)^2$ - 1 = n^2 + 1 + 2n - 1 = n^2 + 2n.

(ii) gofoh (n) = n - 1, gof (n - 1) = $(n-1)^2$
g [$(n-1)^2$] = $(n-1)^2$ + 1 = n^2 + 1 - 2n + 1 = n^2 - 2n + 2.

(iii) fogoh (n) = n - 1

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fog (n - 1) = (n - 1) + 1
f (n) = n^2.

Note:

If f and g are one-to-one, then the function (gof) (gof) is also one-to-one.

If f and g are onto then the function (gof) (gof) is also onto.

Composition consistently holds associative property but does not hold commutative property.

Recursively Defined Functions

Most of the functions we have dealt with in previous chapters have been defined explicitly: by a formula in terms of the variable. We can also define functions recursively: in terms of the same function of a smaller variable. In this way, a recursive function "builds" on itself.

A recursive definition has two parts:

1. Definition of the smallest argument (usually $f(0)$ or $f(1)$).
2. Definition of $f(n)$, given $f(n-1)$, $f(n-2)$, etc.

Here is an example of a recursively defined function:

$$\begin{cases} f(0) = 5 \\ f(n) = f(n-1) + 2 \end{cases}$$

We can calculate the values of this function:

$f(0)=5$
$f(1)=f(0) + 2 = 5 + 2 = 7$
$f(2)=f(1) + 2 = 7 + 2 = 9$
$f(3)=f(2) + 2 = 9 + 2 = 11$
...

This recursively defined function is equivalent to the explicitly defined function $f(n) = 2n + 5$. However, the recursive function is defined only for nonnegative integers.

Here is another example of a recursively defined function:

$$\begin{cases} f(0) = 0 \\ f(n) = f(n-1) + 2n - 1 \end{cases}$$

The values of this function are:

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$f(0)=0$
$f(1)=f(0)+(2)(1)-1=0+2-1=1$
$f(2)=f(1)+(2)(2)-1=1+4-1=4$
$f(3)=f(2)+(2)(3)-1=4+6-1=9$
$f(4)=f(3)+(2)(4)-1=9+8-1=16$
...

This recursively defined function is equivalent to the explicitly defined function $f(n) = n^2$. Again, the recursive function is defined only for nonnegative integers.

Here is one more example of a recursively defined function: $\begin{cases} f(0) = 1 \\ f(n) = n \cdot f(n-1) \end{cases}$

The values of this function are:

$f(0)=1$
$f(1)=1ff(0)=1f1=1$
$f(2)=2ff(1)=2f1=2$
$f(3)=3ff(2)=3f2=6$
$f(4)=4ff(3)=4f6=24$
$f(5)=5ff(4)=5f24=120$
...

This is the recursive definition of the factorial function, $F(n) = n!$.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

POSETS

LECTURE NO:-16

Partially Ordered Sets

Consider a relation R on a set S satisfying the following properties:

1. R is reflexive, i.e., xRx for every $x \in S$.
2. R is antisymmetric, i.e., if xRy and yRx, then x = y.
3. R is transitive, i.e., xRy and yRz, then xRz.

Then R is called a partial order relation, and the set S together with partial order is called a partially order set or POSET and is denoted by $(S, \leq)$.

Example:

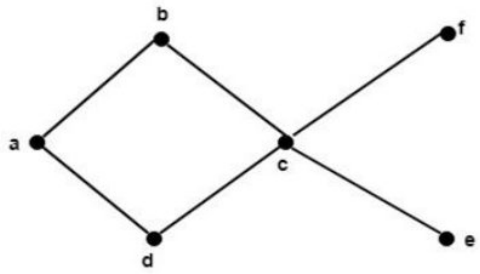
1. The set N of natural numbers form a poset under the relation '$\leq$' because firstly $x \leq x$, secondly, if $x \leq y$ and $y \leq x$, then we have x = y and lastly if $x \leq y$ and $y \leq z$, it implies $x \leq z$ for all $x, y, z \in N$.
2. The set N of natural numbers under divisibility i.e., 'x divides y' forms a poset because x/x for every $x \in N$. Also if x/y and y/x, we have x = y. Again if x/y, y/z we have x/z, for every $x, y, z \in N$.
3. Consider a set S = {1, 2} and power set of S is P(S). The relation of set inclusion $\subseteq$ is a partial order. Since, for any sets A, B, C in P (S), firstly we have $A \subseteq A$, secondly, if $A \subseteq B$ and $B \subseteq A$, then we have A = B. Lastly, if $A \subseteq B$ and $B \subseteq C$,then $A \subseteq C$. Hence, $(P(S), \subseteq)$ is a poset.

Elements of POSET:

1. **Maximal Element:** An element $a \in A$ is called a maximal element of A if there is no element in c in A such that $a \leq c$.
2. **Minimal Element:** An element $b \in A$ is called a minimal element of A if there is no element in c in A such that $c \leq b$

: Determine all the maximal and minimal elements of the poset whose Hasse diagram is shown in fig:

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Solution: The maximal elements are b and f.

The minimal elements are d and e.

Comparable Elements:

Consider an ordered set A. Two elements a and b of set A are called comparable if $a \leq_R b$ or $b \leq_R a$

Non-Comparable Elements:

Consider an ordered set A. Two elements a and b of set A are called non-comparable if neither $a \leq b$ nor $b \leq a$.

Example: Consider A = {1, 2, 3, 5, 6, 10, 15, 30} is ordered by divisibility. Determine all the comparable and non-comparable pairs of elements of A.

Solution: The comparable pairs of elements of A are:

{1, 2}, {1, 3}, {1, 5}, {1, 6}, {1, 10}, {1, 15}, {1, 30}

{2, 6}, {2, 10}, {2, 30}

{3, 6}, {3, 15}, {3, 30}

{5, 10}, {5, 15}, {5, 30}

{6, 30}, {10, 30}, {15, 30}

The non-comparable pair of elements of A are:

{2, 3}, {2, 5}, {2, 15}

{3, 5}, {3, 10}, {5, 6}, {6, 10}, {6, 15}, {10, 15}

Linearly Ordered Set: Toset

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Consider an ordered set A. The set A is called linearly ordered set or totally ordered set, if every pair of elements in A is comparable.

Example: The set of positive integers I_+ with the usual order $\leq$ is a linearly ordered set.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

HASSE DIAGRAMS

LECTURE NO:-17

Hasse Diagrams

A **Hasse diagram** is a *graphical representation* of the relation of elements of a **partially ordered set (poset)** with an implied *upward orientation*. A point is drawn for each element of the partially ordered set (poset) and joined with the line segment according to the following rules:

- If *p<q* in the poset, then the point corresponding to *p appears lower* in the drawing than the point corresponding to q.
- The two points p and q will be joined by line segment *iff p is related to q*.

To draw a Hasse diagram, provided set must be a poset.

A poset or partially ordered set A is a pair, (B,) of a set B whose elements are called the vertices of A and obeys following rules:

1. Reflexivity → p p p B
2. Anti-symmetric → p q and q p iff p=q
3. Transitivity → if p q and q r then p r

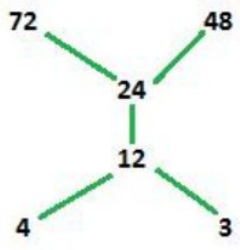
Example-1: Draw Hasse diagram for ({3, 4, 12, 24, 48, 72}, /)

Explanation – According to above given question first, we have to find the poset for the divisibility.

Let the set is A.

A={(3 12), (3 24), (3 48), (3 72), (4 12), (4 24), (4 48), (4 72), (12 24), (12 48), (12 72), (24 48), (24 72)}

So, now the Hasse diagram will be:

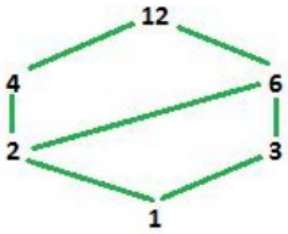


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In above diagram, 3 and 4 are at same level because they are not related to each other and they are smaller than other elements in the set. The next succeeding element for 3 and 4 is 12 i.e, 12 is divisible by both 3 and 4. Then 24 is divisible by 3, 4 and 12. Hence, it is placed above 12. 24 divides both 48 and 72 but 48 does not divide 72. Hence 48 and 72 are not joined.
We can see transitivity in our diagram as the level is increasing.

Example-2: Draw Hasse diagram for (D , /)
Explanation – Here, D means set of positive integers divisors of 12.
So, D ={1, 2, 3, 4, 6, 12}
poset A = {(1 2), (1 3), (1 4), (1 6), (1 12), (2 4), (2 6), (2 12), (3 6), (3 12), (4 12), (6 12)}
So, now the Hasse diagram will be-



In above diagram, 1 is the only element that divides all other elements and smallest. Hence, it is placed at the bottom. Then the elements in our set are 2 and 3 which do not divide each other hence they are placed at same level separately but divisible by 1 (both joined by 1). 4 is divisible by 1 and 2 while 6 is divisible by 1, 2 and 3 hence, 4 is joined by 2 and 6 is joined by 2 and 3. 12 is divisible by all the elements hence, joined by 4 and 6 not by all elements because we have already joined 4 and 6 with smaller elements accordingly.

For regular Hasse Diagram:

- **Maximal element** is an element of a POSET which is not less than any other element of the POSET. Or we can say that it is an element which is not related to any other element. Top elements of the Hasse Diagram.
- Example- In the diagram above, we can say that 4,2,3,6,1 are related to 12 (ordered by division e.g. (4,/)) but 12 is not related to any other. (As Hasse Diagram is upward directional).
- **Minimal element** is an element of a POSET which is not greater than any other element of the POSET. Or we can say that no other element is related to this element. Bottom elements of the Hasse Diagram.
- Example- In the diagram above, we can say that 1 is related to 2,3,4,6,12 (ordered by division e.g. (4,/)) but no element is related to 1. (As Hasse Diagram is upward directional).

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- **Greatest element** (if it exists) is the element succeeding all other elements.
- **Least element** is the element that precedes all other elements.

Note – Greatest and Least element in Hasse diagram are only one.

In Example-1,

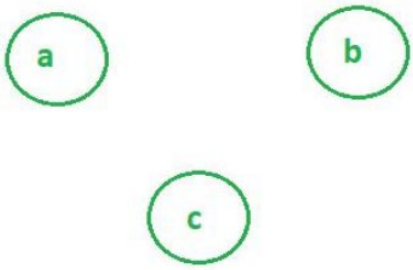
Maximal elements are 48 and 72 since they are succeeding all the elements.
Minimal elements are 3 and 4 since they are preceding all the elements.
Greatest element does not exist since there is no any one element that succeeds all the elements.
Least element does not exist since there is no any one element that precedes all the elements.

In Example-2,

Maximal and Greatest element is 12 and Minimal and Least element is 1.

NOTE: An element can be both maximal and minimal

Example-



Here a,b,c are maximal as well as minimal.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-2 (DISCRETE MATHEMATICAL STRUCTURE)

LATTICES

LECTURE NO:-18

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Lattices:

Let L be a non-empty set closed under two binary operations called meet and join, denoted by $\wedge$ and $\vee$. Then L is called a lattice if the following axioms hold where a, b, c are elements in L:

1) Commutative Law: -

(a) $a \wedge b = b \wedge a$ (b) $a \vee b = b \vee a$

2) Associative Law:-

(a) $(a \wedge b) \wedge c = a \wedge (b \wedge c)$ (b) $(a \vee b) \vee c = a \vee (b \vee c)$

3) Absorption Law:

(a) $a \wedge (a \vee b) = a$ (b) $a \vee (a \wedge b) = a$

Duality:

The dual of any statement in a lattice $(L, \wedge, \vee)$ is defined to be a statement that is obtained by interchanging $\wedge$ an $\vee$.

For example, the dual of $a \wedge (b \vee a) = a \vee a$ is $a \vee (b \wedge a) = a \wedge a$

Bounded Lattices:

A lattice L is called a bounded lattice if it has greatest element 1 and a least element 0.

Example:

1. The power set P(S) of the set S under the operations of intersection and union is a bounded lattice since $\varnothing$ is the least element of P(S) and the set S is the greatest element of P(S).
2. The set of +ve integer I_+ under the usual order of $\leq$ is not a bounded lattice since it has a least element 1 but the greatest element does not exist.

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Properties of Bounded Lattices:

If L is a bounded lattice, then for any element a $\in$ L, we have the following identities:

1. a $\vee$ 1 = 1
2. a $\wedge$ 1= a
3. a $\vee$ 0=a
4. a $\wedge$ 0=0

Theorem: Prove that every finite lattice L = $\{a_1,a_2,a_3....a_n\}$ is bounded.

Proof: We have given the finite lattice:

$L = \{a_1,a_2,a_3....a_n\}$

Thus, the greatest element of Lattices L is $a_1 \vee a_2 \vee a_{3 \vee .... \vee an.}$

Also, the least element of lattice L is $a_1 \wedge a_2 \wedge a_3 \wedge .... \wedge a_n$.

Since, the greatest and least elements exist for every finite lattice. Hence, L is bounded.

Sub-Lattices:

Consider a non-empty subset L_1 of a lattice L. Then L_1 is called a sub-lattice of L if L_1 itself is a lattice i.e., the operation of L i.e., a $\vee$ b $\in$ L_1 and a $\wedge$ b $\in$ L_1 whenever a $\in$ L_1 and b $\in$ L_1.

Example: Consider the lattice of all +ve integers I_+ under the operation of divisibility. The lattice D_n of all divisors of n > 1 is a sub-lattice of I_+.

Determine all the sub-lattices of D_{30} that contain at least four elements, D_{30}={1,2,3,5,6,10,15,30}.

Solution: The sub-lattices of D_{30} that contain at least four elements are as follows:

1. {1, 2, 6, 30}
2. 2. {1, 2, 3, 30}
3. {1, 5, 15, 30}
4. 4. {1, 3, 6, 30}
5. {1, 5, 10, 30}
6. 6. {1, 3, 15, 30}
7. {2, 6, 10, 30}

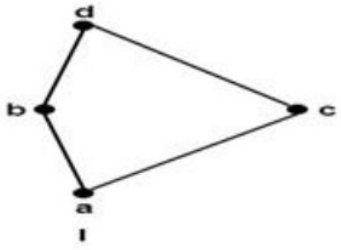
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Isomorphic Lattices:

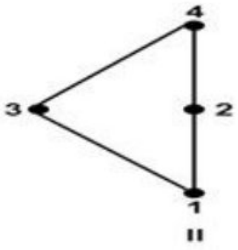
Two lattices L_1 and L_2 are called isomorphic lattices if there is a bijection from L_1 to L_2 i.e., f: $L_1 \longrightarrow L_2$, such that f (a $\wedge$ b) =f(a)$\wedge$ f(b) and f (a $\vee$ b) = f (a) $\vee$ f (b)

Example: Determine whether the lattices shown in fig are isomorphic.

Solution: The lattices shown in fig are isomorphic. Consider the mapping f = {(a, 1), (b, 2), (c, 3), (d, 4)}.For example f (b $\wedge$ c) = f (a) = 1. Also, we have f (b) $\wedge$ f(c) = 2 $\wedge$ 3 = 1



I



II

Distributive Lattice:

A lattice L is called distributive lattice if for any elements a, b and c of L,it satisfies following distributive properties:

1. a $\wedge$ (b $\vee$ c) = (a $\wedge$ b) $\vee$ (a $\wedge$ c)
2. a $\vee$ (b $\wedge$ c) = (a $\vee$ b) $\wedge$ (a $\vee$ c)

If the lattice L does not satisfies the above properties, it is called a non-distributive lattice.

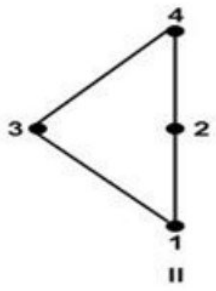
Example:

1. The power set P (S) of the set S under the operation of intersection and union is a distributive function. Since,

   a $\cap$ (b $\cup$ c) = (a $\cap$ b) $\cup$ (a $\cap$ c) and, also a $\cup$ (b $\cap$ c) = (a $\cup$ b) $\cap$ (a $\cup$c) for any sets a, b and c of P(S).

2. The lattice shown in fig II is a distributive. Since, it satisfies the distributive properties for all ordered triples which are taken from 1, 2, 3, and 4.

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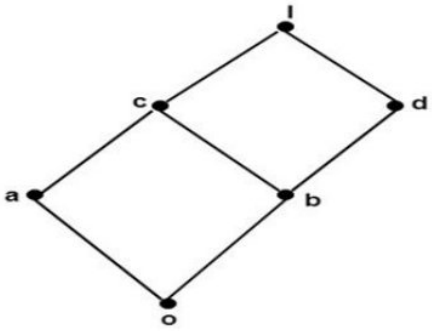


Complements and complemented lattices:

Let L be a bounded lattice with lower bound o and upper bound I. Let a be an element if L. An element x in L is called a complement of a if a $\vee$ x = I and a $\wedge$ x = 0

A lattice L is said to be complemented if L is bounded and every element in L has a complement.

Example: Determine the complement of a and c in fig:



Solution: The complement of a is d. Since, a $\vee$ d = 1 and a $\wedge$ d = 0

The complement of c does not exist. Since, there does not exist any element c such that c $\vee$ c'=1 and c $\wedge$ c'= 0.

Modular Lattice:

A lattice (L, $\wedge, \vee$) is called a modular lattice if a $\vee$ (b $\wedge$ c) = (a $\vee$ b) $\wedge$ c whenever a $\leq$ c.

Direct Product of Lattices:

Let (L_1 $\vee_1$ $\wedge_1$)and (L_2 $\vee_2$ $\wedge_2$) be two lattices. Then (L, $\wedge, \vee$) is the direct product of lattices, where L = L_1 x L_2 in which the binary operation $\vee$(join) and $\wedge$(meet) on L are such that for any $(a_1,b_1)$and (a_2,b_2) in L.

$(a_1,b_1) \vee (a_2,b_2) = (a_1 \vee_1 a_2, b_1 \vee_2 b_2)$

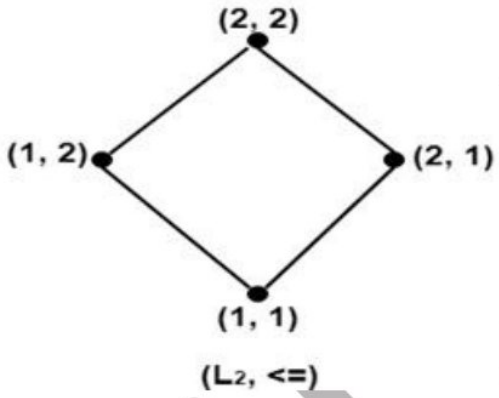
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and $(a_1,b_1) \wedge (a_2,b_2) = (a_1 \wedge_1 a_2, b_1 \wedge_2 b_2)$.

Example: Consider a lattice (L, ≤) as shown in fig. where L = {1, 2}. Determine the lattices $(L^2, \leq)$, where $L^2 = L \times L$.



Solution: The lattice $(L^2, \leq)$ is shown in fig:



References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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